

Evidence-gated online GKP decoding: parallel multimode qualification and deterministic fail-closed RTL

Phase 6D evidence cutoff: 21 July 2026

Abstract

Online Gottesman–Kitaev–Preskill (GKP) decoding couples two questions that need not have the same answer: whether an observed-only adaptive decoder improves logical error rate under drift, and whether a decoder action can be delivered with deterministic and safe digital semantics. We separate these questions into parallel evidence lanes joined by an interface contract rather than a common performance denominator. The multimode software lane registers exact logical-coset maximum-likelihood decoding, static mixture models and causal adaptive frontends under a common observation and budget contract. Its development-only entry test found identical logical error rates for the strongest retained static-mixture exact-MLD baseline and the proposed causal risk action, $p_L = 0.111979$ over 79,872 rounds, giving a paired relative improvement and 95% lower confidence bound of 0%. The registered pilot, formal benchmark and scaling study were therefore not accessed: multimode frozen-benchmark SOTA is not established.

Independently, the exact single-mode RTL lane integrates the MAP fast path, event policy, admission logic and versioned A/B parameter banks in one production top. It passes 17/17 formal gates and kills 21/21 targeted formal mutations. An independent one-million-cycle CXXRTL replay compares the full 148-byte public output vector with zero mismatch, zero undefined action and zero silent overflow while preserving a six-cycle source-to-action contract at initiation interval one. Three open-source placement-and-routing seeds meet the 27-MHz contract clock; these are whole-harness pre-board estimates, not physical measurements. Board latency, jitter, deadline misses and power remain null, and no FPGA speed advantage is claimed. CNN and student models are absent from the primary result and remain replaceable, dropped approximation modules. The resulting contribution is a falsifiable qualification method plus a deterministic, atomic and fail-closed pre-board implementation, without transferring evidence between the two lanes.

1 Introduction

GKP codes encode finite-dimensional information in oscillator phase space and turn small displacement errors into analog modular syndromes [1]. Analog likelihoods can materially improve decoding when the noise model is known [2, 3], while multimode lattice geometry creates a separate algorithmic problem whose solution can range from closest point decoding [4] to exact logical-coset maximum likelihood and its K -matching approximations [5, 6]. A deployed decoder, however, must operate while calibration, correlation and fault statistics change. It must also produce actions within a fixed control deadline and prevent partially validated parameters from reaching the fast path.

These demands invite an attractive but unsafe narrative: one adaptive model predicts drift, a neural network improves logical error rate, and an FPGA makes the resulting decoder fast. Each clause uses a different task signature. Logical-error evidence depends on the code, channel, syndrome history, available actions, comparator and statistical unit. Hardware evidence depends on the implemented top, precision, source-to-action boundary, initiation interval, update transaction and physical transport. A positive result in one signature cannot supply a missing denominator in another. In particular, short latency in a single-mode lookup path cannot establish multimode decoding accuracy, and a software LER result cannot prove an RTL safety property.

We therefore formulate an evidence-gated dual-lane system. The multimode software lane asks whether observed-only, causal, posterior-aware actions have prospective headroom over the strongest deployable static decoder. Its registered baseline set includes Euclidean and weighted closest-point decoding, periodic analog-likelihood matching, exact logical-coset MLD [5], K -minimum-weight matchings [6], a syndrome-estimated static mixture exact MLD, and causal Window, EWMA, Kalman, SMC and change-point frontends. The first question is deliberately cheaper than a full benchmark: if the best causal action cannot improve the strongest retained baseline on a fresh development split, more estimator complexity cannot be promoted merely by being trained.

The second lane asks a narrower but constructive hardware question. Given an exact single-mode MAP/event design, can one actual RTL top guarantee a six-cycle, initiation-interval-one action path while accepting parameter updates atomically and failing closed on invalid data? This lane uses formal properties, mutation closure, an independent long-sequence golden model and whole-harness post-route estimates. It does not claim that the current RTL executes the multimode MLD backend, and it reserves all physical-board fields for a later named-board experiment. No LER—latency weighted score is defined.

The two contributions are thus parallel. First, we provide a source-grounded multimode benchmark and a causal-headroom gate that reaches a negative decision before pilot or formal data can be inspected. Second, we provide an exact single-mode production RTL whose timing and transaction properties are qualified pre-board. A shared packet, image and evidence contract explains how a future multimode compiler could target the same publication mechanism, but it is not an equivalence proof or a current deployment arrow. CNN, teacher and student components have no independent vote: they can approximate an authorized posterior, likelihood or action only after retention and budget gates are passed.

This organization changes the paper’s claim from “a CNN–FPGA decoder wins” to “a candidate algorithm is allowed to fail its strongest-baseline gate, while a distinct deterministic hardware contract is proved on the exact top that exists.” That separation is the main methodological result. It also makes the remaining scientific target unambiguous: a future multimode method must exceed every eligible deployable baseline with a simultaneous paired lower confidence bound above the registered 10% threshold, rather than using hardware safety or a weaker comparator as evidence of LER superiority.

2 Related work and task signatures

2.1 Static, analog and multimode decoding

Closest-integer binning is a useful low-cost anchor, but it discards analog reliability. Analog matching retains likelihood information [2], and circuit-level surface–GKP studies add repeated noisy measurements and space–time structure [3]. Multimode closest-point decoding instead treats a lattice geometry problem [4]. These methods are not interchangeable rows: code family, measurement history and decoder objective must match before LER values are ranked. In our target signature, static-mixture exact logical-coset MLD is the strongest retained deployable baseline because it marginalizes a frozen syndrome-estimated prior at the logical-coset level.

2.2 Adaptation, learned control and physical lifetime

Syndrome statistics can estimate noise without privileged access to the latent error [7]. Online Window, EWMA, Kalman, particle and change-point methods differ in delay, memory and their ability to represent abrupt transitions. Offline decoder-prior calibration on experimental data is a valuable but different task [8]. Non-Markovian feedback and learned controllers can optimize a physical protocol when their observation, action and wall-clock budgets are defined [9]. Likewise, real-time QEC beyond break-even is a physical lifetime claim requiring a matched device experiment [10]. None of those results supplies a same-task numeric denominator for the present syndrome-only multimode study, and our per-round simulation does not estimate a physical lifetime.

2.3 FPGA decoders

Hardware QEC decoders such as LILLIPUT, Helios and collision-clustering designs demonstrate important latency and scaling strategies for other code families [11, 12, 13]. Raw nanoseconds cannot be ranked across different syndrome sources, distances, precisions, timing boundaries or boards. Our hardware result therefore reports cycles, initiation interval, formal scope and whole-harness timing closure. It makes no claim to be faster than an existing FPGA decoder.

3 Methods

3.1 Two-lane evidence contract

An evidence lane is a tuple

$$\mathcal{T} = (\mathcal{C}, \mathcal{N}, \mathcal{O}, \mathcal{A}, \mathcal{D}, \mathcal{P}, \mathcal{B}, \mathcal{M}, \mathcal{S}), \quad (1)$$

where $\mathcal{C}$ is the code family, $\mathcal{N}$ the channel, $\mathcal{O}$ the online observations, $\mathcal{A}$ the action space, $\mathcal{D}$ the decoder backend, $\mathcal{P}$ the precision, $\mathcal{B}$ the compute budget, $\mathcal{M}$ the metric namespace and $\mathcal{S}$ the statistical unit. A claim may move between methods only when these fields match or when an explicit adapter and its error are qualified. The software and RTL lanes share versioned packet/image interfaces and fail-closed publication semantics; they do not share $\mathcal{C}$, $\mathcal{D}$, $\mathcal{M}$ or $\mathcal{S}$.

Three transfers are forbidden throughout this work. An RTL property cannot rescue a failed multimode LER gate. A multimode simulation cannot validate an RTL timing or safety property. A learned approximation cannot alter either primary verdict unless its own matched retention, precision and resource gates are passed. Consequently, no LER–latency weighted score is defined.

3.2 Multimode software qualification lane

The deployable software record contains quantized syndromes and health fields available up to the current decision. Latent drift state, scenario identity, future samples and logical labels are isolated in an evaluator record. Prefix, future-suffix and truth-mutation tests check that the deployable path is observed-only. The exact logical-coset backend computes

$$\hat{L}_t = \arg \max_L \int p(L, s_t | \theta) p(\theta | s_{<t}) d\theta, \quad (2)$$

with a point prior recovering plug-in exact MLD. A static mixture freezes $p(\theta)$ from the allowed pre-formal data and is therefore deployable.

The registered development entry test uses 12 seeds, 13 drift families and 512 rounds per family, or 79,872 paired rounds per method. All methods see the same trace. Baseline selection, causal ceiling and action-value decomposition are computed before any pilot or formal access. Continuation requires at least 15% point headroom and a 12% paired 95% lower confidence bound over the strongest retained development baseline. Failure stops implementation, pilot, formal and scaling tasks. The later frozen-benchmark promotion rule is stricter: the simultaneous paired lower bound must exceed 10% against every eligible deployable baseline, with a positive absolute lower bound and consistent $d = 3, 5$ behavior at registered noise points.

We also preserve an earlier opened task-local multimode experiment. It uses a different denominator and was inspected before the Phase 6D split; hence its LER, tail and host-runtime values are descriptive context, not a ranking result and not evidence for the registered promotion gate.

3.3 Exact single-mode RTL lane

The production top instantiates the parameter-bank manager, commit admission, five-stage MAP core and one-stage event/action policy exactly once. There is no raw configuration or trust bypass into the core. The source-to-action contract is

$$t_{\text{action}} - t_{\text{source}} = 6 \text{ cycles}, \quad \Pi = 1. \quad (3)$$

The contract ends at the registered RTL output and excludes physical transport, CDC, ADC/DAC and control electronics.

Each update stages a complete image in the inactive bank. Admission checks image completeness, CRC, version, expected active bank and trust state. A compare-and-swap commit occurs only at a safe boundary; the retired bank is protected for the six-cycle pipeline drain. Integrity, age, leakage, posterior-sum or version faults select a defined fallback and can republish the last-known-good image. Image generation and activation epochs are distinct, so a trusted older image may be restored without pretending that its payload was regenerated.

Formal qualification covers all-state management/admission guards, transition refinement, bounded reachability, the actual core’s fail-closed output and three-step atomic commit behavior. Cover traces exercise activation, LKG recovery, cancellation, drain rejection, near-wrap rejection, host conflict and policy priority. Targeted semantic mutations remove or corrupt one protection at a time; a property suite is accepted only if every registered mutation is killed.

The independent long-sequence model reimplements admission, management, core and policy behavior without importing the RTL implementation. It compares a 148-byte public vector cycle by cycle over ten 100,000-cycle families. The trace includes full-image transactions, host/policy commit races, rejected updates, integrity faults, leakage, recovery and compound transport events. Post-route qualification uses the same whole-harness observability top and three fixed seeds. Frequency and resources are estimates from the open-source flow; no vendor timing signoff or physical-board measurement is inferred.

3.4 Replaceable CNN/student extension

A learned module is permitted only as an approximation to an authorized posterior, log-likelihood ratio, logical-coset probability or action. It must use the same train/calibration/pilot/formal split and online information as its teacher, and must report calibration, action agreement, LER retention, worst-family retention, runtime, memory and quantization error under a matched budget. Phase 6D did not authorize the teacher because the multimode entry test stopped before implementation. CNN/student training, quantization and formal retention were therefore not run, and legacy CNN evidence cannot reopen either primary lane.

3.5 Evidence states and reproducibility

We use distinct states for measured, estimated, reproduced, contextual, negative, dropped, blocked and not-applicable values. Null is not zero. Every primary statement is bound to a report, raw data or trace, configuration, implementation, source locator and SHA-256 hash. Negative decisions retain their denominator and unavailable downstream outputs. The machine contract rejects mutations that delete the strongest baseline, fill a board-null field, promote learning, merge lane metrics or reuse a historical figure as current evidence.

4 Results

4.1 The multimode causal-headroom gate is negative

The strongest development baseline was static-mixture exact MLD. It and the best causal risk action each made 8,944 logical errors in 79,872 paired rounds, giving

$$p_L^{\text{static mixture MLD}} = p_L^{\text{causal risk}} = 0.1119791667. \quad (4)$$

The paired relative improvement was 0%, with a 95% lower confidence bound of 0%. Both the 15% point and 12% lower-bound entry thresholds failed. The verdict was therefore `NO_GO_MULTIMODE_CAUSAL_HEADROOM`; no proposed posterior-predictive implementation was selected, and pilot, formal and distance/noise scaling were not accessed.

The decomposition explains why additional estimator capacity was not a promising rescue. Relative to intermediate variants, the observed-only estimator, metric, coset, marginalization and action components contributed -0.001277 , -0.033654 , -0.014636 , -0.000013 and $+0.055551$ absolute LER, respectively. The trusted action intervened 9,884 times and removed 4,437 errors relative to the unprotected action variant, but only returned the system to the static-mixture exact-MLD result. The action policy repaired errors created upstream; it did not create net headroom over the strongest static decoder.

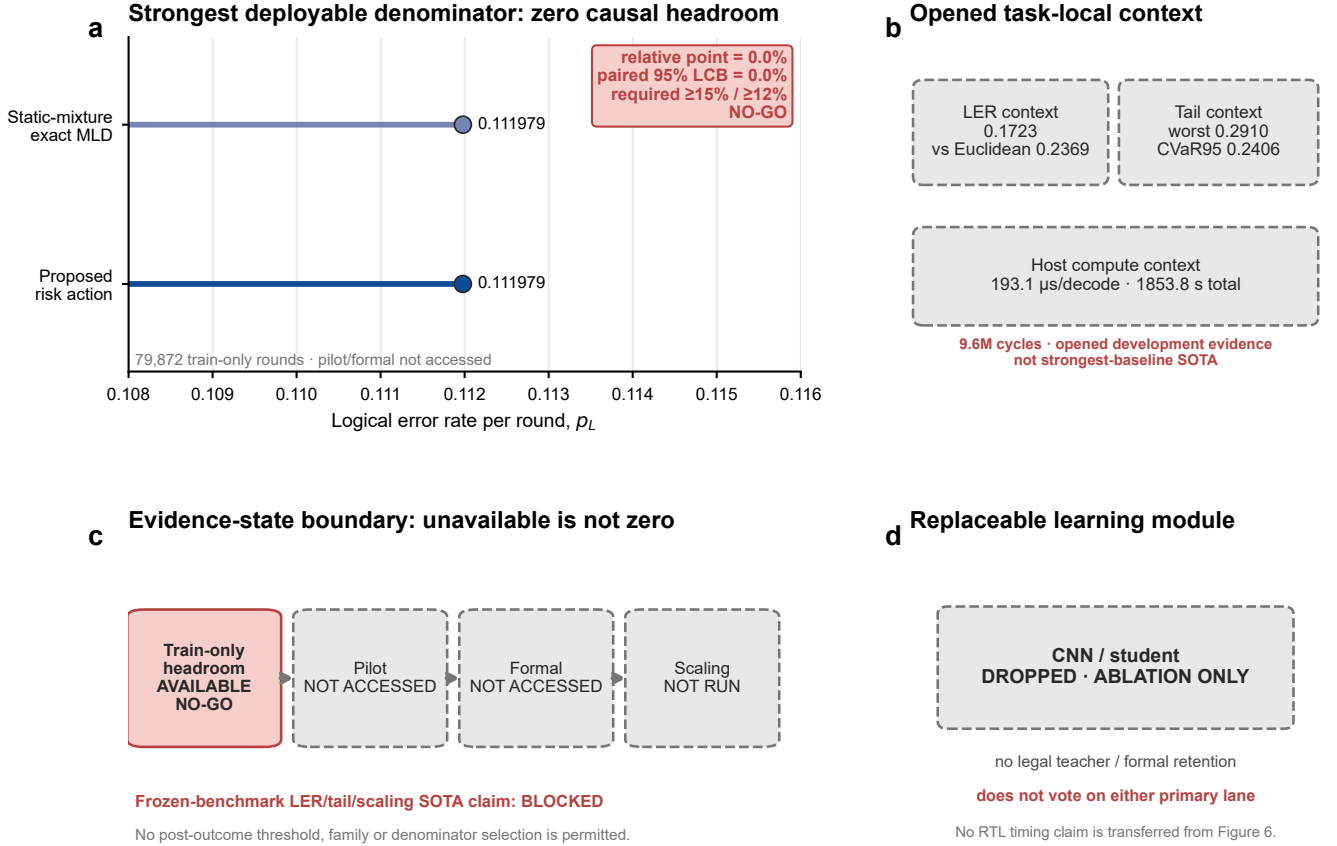


Figure 1: Multimode software evidence. (a) The strongest-baseline entry comparison has zero point and lower-bound headroom. (b) An opened $d = 3$ task-local study is retained only as LER, tail and host-compute context because it does not use the strongest denominator. (c) Pilot, formal and scaling evidence remain not run after the entry stop. (d) CNN/student is a dropped ablation state. No RTL result is used to alter this verdict.

4.2 Opened multimode results remain contextual

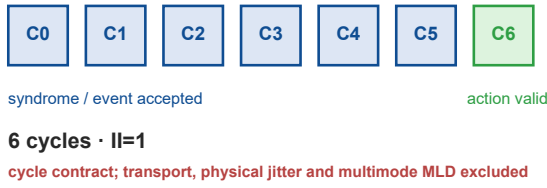
The earlier opened $d = 3$ study processed 9.6 million physical cycles across 32 seed clusters with 20,000 bootstrap resamples. The candidate had $p_L = 0.17226094$ versus 0.23692938 for static Euclidean decoding, an absolute difference of 0.06466844 with interval $[0.06441281, 0.06492636]$. Its non-overlapping 512-cycle worst-window LER was 0.291015625, CVar95 was 0.24062667, and measured host runtime was 1853.802 s. These values show that the simulator and tail accounting are operational. They do not establish SOTA because static Euclidean decoding is weaker than the subsequently retained static-mixture exact MLD and the data belong to an opened, task-local context.

4.3 The exact single-mode RTL passes deterministic and safety gates

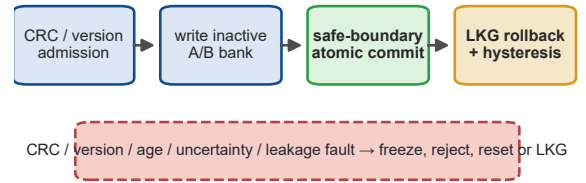
All 17 registered formal gates passed, and all 21 targeted formal mutations were killed. The proof scope includes inductive safety invariants and transition refinement for the converged top, bounded reachability, and direct properties on the actual MAP core. It does not include unbounded liveness, physical CDC/metastability or a multimode decoder.

The independent CXXRTL qualification processed 1,000,000 cycles. Of these, 998,462 had valid inputs and outputs. All 998,435 adjacent valid input pairs produced adjacent outputs, confirming $\Pi=1$. The observed source-to-action latency was exactly six cycles and MAP latency exactly five cycles. Full-vector comparison found zero mismatch, zero latency violation, zero undefined action, zero CRC error and zero silent version wrap or arithmetic overflow. The run included 4 completed bank commits, 28 management rejects, 471 fault-to-clean recoveries and 882 route-to-open recoveries rather than a nominal-only trace.

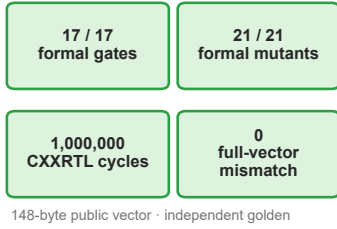
a Deterministic source-to-action pipeline



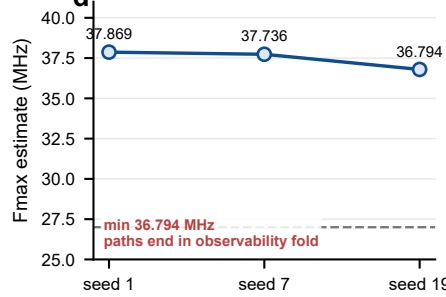
b Atomic A/B bank and fail-closed recovery



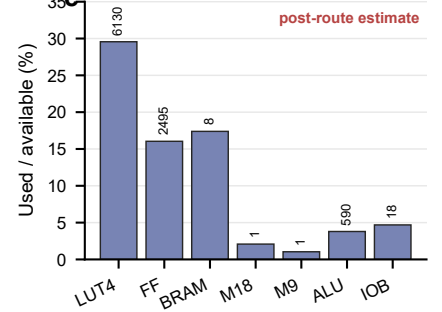
c Exact qualification



d Three-seed whole-harness P&R



e Complete-harness resources



f Physical-board and speed-claim boundary



Figure 2: Exact single-mode pre-board RTL evidence. (a) The source-to-action contract is six cycles at $\Pi=1$. (b) CRC/version admission, inactive-bank staging, safe-boundary commit and LKG recovery implement atomic fail-closed publication. (c) Formal and one-million-cycle CXXRTL qualification pass. (d,e) Three post-route seeds meet 27 MHz for the whole observability harness; resources and frequency are estimates. (f) Board latency, jitter, deadline misses, power and physical transaction latency remain null. The current RTL does not execute the multimode decoder.

All three placement-and-routing seeds met the 27-MHz contract clock. Estimated Fmax ranged from 36.794 to 37.869 MHz with a median of 37.736 MHz. Median whole-harness resources were 6,118 LUT4, 2,495 DFF, 8 BSRAM, one 18x18 multiplier and one 9x9 multiplier. Critical paths terminated in the observability

fold, so the frequency is not relabelled as a bare-core result. The six-cycle clock model corresponds to 222.222 ns at 27 MHz and 163.068 ns at minimum estimated Fmax, but neither number includes transport or physical jitter and neither is a board measurement.

4.4 The combined verdict is RTL-only

The two Boolean lane decisions are multimode=false and RTL=true. The resulting publication verdict is GO_RTL_ONLY: the exact single-mode deterministic/atomic/fail-closed pre-board contribution is open, while multimode SOTA, learning promotion, board measurement and FPGA speed ranking remain closed. No scalar combines LER with cycles or resources. The positive RTL lane neither compensates for nor weakens the mandatory multimode negative.

5 Discussion

The principal lesson is that algorithmic adaptivity and safe deterministic execution are separable scientific claims. The multimode result is informative precisely because it prevents a large engineering effort after a strong static model erased the apparent causal gain. Against static Euclidean decoding the opened study looked favorable; against static-mixture exact MLD the registered causal ceiling had zero headroom. This is consistent with a regime in which past syndromes identify enough of a stationary mixture for a frozen marginal decoder, while real-time state estimates are too noisy or too delayed to change the logical action profitably.

The decomposition also clarifies what must change before a renewed multimode claim is credible. A future proposal needs new causal information or a new action space, not simply a larger predictor. Candidate directions include sensor-side calibration variables that are genuinely available online, longer-horizon posterior prediction validated prequentially, or physical control actions whose counterfactual value exceeds the static decoder. Each would require a fresh immutable split and the same strongest baseline set. It would not be permissible to tune thresholds separately on calibration and telegraph families after seeing the formal result.

The RTL contribution solves a different failure mode. An estimator can be statistically useful and still publish a torn, stale or untrusted image, miss a deadline, or return an undefined action. The converged top makes the safety authority explicit: full-image staging, versioned compare-and-swap, pipeline drain and LKG recovery are part of the decoder interface. Formal mutation and long-sequence fault traffic provide stronger pre-board evidence than a nominal demo. Nevertheless, fail closed is a digital invariant, not a claim that the fallback action minimizes LER or that the physical system is safe.

The contract bridge is consequently architectural rather than empirical. A future multimode compiler may target versioned images and the same transaction protocol, but it must first prove quantization and action equivalence, meet the six-cycle/ $\Pi=1$ or a newly registered timing contract, and repeat formal, mutation, long-sequence and post-route qualification on the new actual top. Until then the current hardware remains exact single mode. This boundary is what prevents two individually valid results from becoming one invalid global claim.

Learning remains optional by construction. A CNN or student could reduce posterior cost or approximate exact MLD if it preserved calibration, actions, LER and worst-family behavior under a matched resource budget. Phase 6D contains no authorized teacher or formal retention result, so making learning central would add a third unsupported claim rather than strengthen the two lanes. The appropriate current use is an explicit dropped ablation and a future replaceable module.

6 Limitations

The multimode experiment is simulation-based. The registered development headroom result covers the frozen $d = 3$ task signature but did not continue to the $d = 3, 5$ pilot/formal/scaling matrix. It does not establish a universal impossibility of adaptive GKP decoding. Some source-backed comparators were

registered but remained ineligible or unexecuted after the stop, including periodic analog-likelihood MWPM, K -MWM, source-transcribed noisy-auxiliary decoding, SMC, BOCPD and IMM. The correct conclusion is a Phase 6D v1 no-go, not that static MLD always dominates adaptation.

The hardware scope is a two-state, exact single-mode digital top. Formal and CXXRTL evidence exclude physical transport, pin constraints, CDC, metastability, analog acquisition, control electronics and the quantum device. Placement and routing use an open-source flow rather than vendor signoff. The reported 27-MHz clock and $F_{\max}$ values are pre-board estimates. Physical-board latency, jitter, deadline misses, power, transfer latency and commit latency are unmeasured and stored as null. We therefore claim neither measured speed nor an advantage over existing FPGA QEC decoders.

Finally, per-round simulated LER cannot be compared with the logical lifetime or break-even values of a cavity experiment or a non-Markovian feedback controller. The present work contains no calibrated cavity-transmon digital twin, real GKP record, closed-loop quantum-device action, measured logical lifetime or physical break-even result. Those claims require a separate protocol-matched experiment.

7 Reproducibility and evidence availability

Figure 1 and Figure 2 are generated from a 94-row lossless Source Data contract that binds 13 figure elements, 10 final claims, 11 read-only historical snapshots and 58 live artifacts. The figure contract passes 22/22 gates and 22/22 semantic mutations. The multimode headroom report stores all 79,872 paired decisions and the component decomposition. The RTL package stores formal commands and outputs, mutation results, the 202-MB long trace, the independent golden output, all 148 public bytes per cycle, synthesis/netlist logs and every placement-and-routing seed.

The canonical current manuscript is this file. The earlier 51-page contract-centric note and T7.1.1–T7.2.5 contracts remain immutable historical snapshots; their V4 or Phase 6C claims do not override the Phase 6D final gate. The Supplement below provides the delta needed to interpret those snapshots.

8 Conclusion

We present two parallel results without combining their denominators. The multimode software lane contributes a strong-baseline registry and a prospective causal-headroom gate; its v1 candidate has 0% improvement and 0% paired lower-bound headroom over static-mixture exact MLD, so frozen-benchmark LER SOTA is not established. The exact single-mode hardware lane contributes a six-cycle, $\Pi=1$ converged RTL that passes formal mutation closure, one-million-cycle independent full-vector equivalence and three-seed whole-harness timing closure, with atomic A/B publication and fail-closed LKG recovery.

The evidence boundary is part of the result. The current RTL does not execute the multimode MLD, learning is dropped to an optional approximation, and all physical-board fields remain null. Thus this work establishes deterministic, atomic and fail-closed pre-board execution for the exact single-mode design, not multimode SOTA, physical lifetime gain or fastest-FPGA status. The next positive algorithmic paper must first create and prospectively demonstrate causal action-value headroom against the same strongest deployable baseline.

A Supplementary delta: terminology and evidence states

Table 1: Canonical terminology for the Phase 6D manuscript.

Term	Meaning	Excluded interpretation
Multimode software lane	Registered simulation task for algorithmic LER qualification	Not implemented in current RTL
Exact single-mode RTL lane	Two-state production top qualified for cycles and digital safety	Not a multimode or physical-device decoder
Static-mixture exact MLD	Frozen syndrome-estimated prior marginalized at logical-coset level	Not a truth-privileged oracle
Opened task-local context	Inspected result with a weaker/different denominator	Not frozen-benchmark evidence
Frozen-benchmark SOTA	Simultaneous $> 10\%$ paired LCB against every eligible baseline	Not reached in Phase 6D
Six-cycle, $\Pi=1$	Registered RTL source-to-action cycle contract	Not measured nanoseconds including transport
Whole-harness post-route estimate	Timing/resource result for the observability top	Not bare-core, vendor-signoff or board measurement
Board-null	Measurement was not performed and the value is null	Not zero latency, jitter, deadline misses or power
CNN/student dropped ablation	Optional approximation without an authorized teacher/formal retention	Not a primary contribution

B Supplementary delta: baseline eligibility

Table 2: Multimode baseline registry and Phase 6D disposition.

Method	Role	Phase 6D state	Ranking boundary
Closest integer	Weak sanity anchor	Registered	Never strongest
Euclidean CPD/MWPM	Official geometric baseline	Context	Same-task only
Weighted CPD	Static/adaptive metric adapter	Registered	Adapter, not source-exact
Periodic analog MWPM	Folded-likelihood comparator	Dropped after stop	Cannot be replaced by Euclidean weights

Method	Role	Phase 6D state	Ranking boundary
Exact logical-coset MLD	Strong backend	Registered	Explicit-coset checks required
K -MWM	Accuracy-cost approximation	Dropped after stop	Full K -LER-cost curve required
Static-mixture exact MLD	Strongest frozen static candidate	Executed in entry test	Primary development denominator
Delayed Window	Causal adaptive frontend	Registered	No future-window samples
EWMA/Kalman	Causal adaptive frontends	Registered	Common calibration and cadence
SMC-EAP	Adapted posterior baseline	Dropped after stop	Must be labelled adapted
BOCPD/IMM	Abrupt-drift baselines	Dropped after stop	Hazard/states frozen on calibration
True- θ exact MLD	Evaluation upper bound	Not ranked	Truth privilege prohibited online

Roy-Pousset-Royer noisy-auxiliary decoding, repeated circuit-level surface-GKP decoding, Puviani NMF, AQEC, experimental prior calibration and existing FPGA decoders remain separate protocol lanes. They may provide mechanistic context but cannot enter a numeric global leaderboard without an exact observation/action/environment/budget match.

C Supplementary delta: complete claim boundary

Table 3: Final claims consumed by the manuscript.

Claim	State	Current evidence	Required boundary
Opened multimode gain	Context only	9.6M-cycle task-local study	Not strongest denominator
Multimode causal headroom	Mandatory negative	0% point and LCB vs static-mixture exact MLD	Pilot/formal unaccessed
Frozen-benchmark SOTA	Blocked	No formal result or simultaneous > 10% LCB	Must remain unclaimed
Deterministic RTL	Restricted positive	Six cycles, II=1, 1M-cycle full-vector replay	Exact single-mode pre-board only
Atomic/fail-closed RTL	Restricted positive	17/17 formal gates; 21/21 mutants	Digital/two-state scope
Post-route estimate	Restricted positive	Three seeds; minimum Fmax 36.794 MHz	Whole harness; no vendor signoff
Board measurement	Blocked	All physical fields null	No measured timing/power
FPGA speed advantage	Prohibited positive	No same-task board comparator	No fastest/SOTA wording

Claim	State	Current evidence	Required boundary
CNN/student	Dropped ablation	No teacher, checkpoint or retention result	No vote in either lane
Dual-lane relation	Mandatory boundary	Global weighted score is null	Contract bridge only

D Supplementary delta: revocation and negative-result audit

The manuscript contract is revoked if any of the following occurs: the static-mixture exact-MLD row is deleted; the 0% headroom is omitted; the opened result is called formal or SOTA; pilot/scaling nulls are filled with zeros; the current RTL is said to run multimode MLD; a post-route clock model is described as board measured; a physical deadline or power value is invented; CNN/student is made primary; a common LER-latency score is added; or one lane is allowed to satisfy the other lane’s gate.

The negative result has complete denominators rather than an absence-only assertion: 12 seeds, 13 families, 512 rounds, 79,872 total paired rounds, a named strongest baseline, exact error counts, paired point and lower-bound effects, registered continuation thresholds and an explicit list of unaccessed downstream stages. The RTL positive result likewise includes non-nominal traffic, rejected transactions, recovery events, every public output byte, three place-and-route seeds and a stated physical exclusion surface.

E Supplementary delta: historical snapshot policy

T7.1.1–T7.1.4 and T7.2.1–T7.2.5 are read-only Phase 6C snapshots. They remain useful for provenance, earlier V4 negative results, method definitions and external-comparison source locators. Their restricted manuscript verdict cannot override the later Phase 6D truth table. Figures 5 and 6 use new output paths and do not overwrite Figures 1–4 or Supplementary Figures S1–S5.

For release, every historical snapshot is identified by path and SHA-256 hash, and the current manuscript contract verifies all live Phase 6D parents. A source, configuration, report or code hash drift invalidates the generated delta until the corresponding experiment and claim gate are rerun.

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